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Linear Algebra — Exercise Workbook

5 November 2026 · Răzvan

How to use this workbook

All exercises from the seven official seminar sheets (*Algebra-S01 … S07*, MS Teams → Algebra-CS-AI → Class Materials), numbered exactly as on the sheets. The theory follows Prof. Crivei's course (C01–C06) and his book *Basic Linear Algebra* (2022).

Tick an exercise once you can solve it on paper without looking at the solution.

Notation (as in the course). K is always a field. $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\mathbb{N}^* = \mathbb{N} \setminus \{0\}$. $S \leq V$ means "S is a subspace of V", $\langle X \rangle$ is the subspace generated by X, $\widehat{a}$ is the class of a in $\mathbb{Z}_n$. Vectors of K^n are written as rows $(x_1, \dots, x_n)$.

Required solution format: the seminar sheets give no explicit format. The solutions document follows the course: name the theorem you use (e.g. "by the characterization of subspaces"), check its conditions one by one, and write the conclusion on its own line. In exam proofs, every step needs a reason.

Exam: partial exam 1 (Chapters 1–2 of the course) on Saturday, 21 November 2026; partial exam 2 (Chapters 3–4) on Saturday, 16 January 2027. Full solutions, the theory with complete proofs and an exam-proof sheet are in *Linear Algebra — Solutions & Theory*.

Extra practice: Chapter 8 has the true/false quizzes from Crivei's book (50 statements) and Chapter 9 has proof practice — the course theorems to reproduce and 13 new exam-style proof exercises. Do them after the seminar exercises, in the last week before the partial exam.

Method finder

If the exercise asks…	Try first	Chapter
"is $*$ an operation on A?"	check that $x * y \in A$ for *all* $x, y \in A$	1
"is $(A, *)$ a monoid / group?"	associativity, identity, inverses — in this order	1, 3
"count operations / relations on a finite set"	count the free cells of the table, use $\lvert B^A \rvert = \lvert B \rvert^{\lvert A \rvert}$	1, 2
"is r an equivalence relation?"	check (r), (s), (t); for "no", give one counterexample	2
"determine the quotient set"	compute the class $r\langle x \rangle$ of a general x	2
"is h a function?"	every x in the domain has *exactly one* image	2
"H is a subgroup / subring / subfield"	characterization theorems: $1 \in H$, $xy^{-1} \in H$ (rings: $x - y$, xy)	3

"f is a homomorphism / isomorphism"	check $f(xy) = f(x)f(y)$; for isomorphism also bijectivity	3
"V is a vector space"	$(V, +)$ abelian group + axioms (L1)–(L4)	4
"is S a subspace?"	$0 \in S$ and $k_1v_1 + k_2v_2 \in S$; for "no", break one condition	4
"write S as a generated subspace"	solve the equations, express the general vector, split by free parameters	5
"$V = S \oplus T$"	$S + T = V$ and $S \cap T = \{0\}$	5
"f is linear / an endomorphism"	$f(k_1v_1 + k_2v_2) = k_1f(v_1) + k_2f(v_2)$; quick test: $f(0) = 0$	5
"Ker f, Im f"	solve $f(v) = 0$; Im $f = \langle f(\text{basis}) \rangle$	5, 7
"linearly (in)dependent"	n vectors in K^n: determinant; otherwise a homogeneous system or rank	6
"basis / coordinates"	n independent vectors in an n-dimensional space; solve a linear system	6
"basis and dimension of a subspace"	parametrize, take the generators, check independence	7
"complement / complete to a basis"	add vectors keeping the determinant $\neq 0$	7
"dimension of $S \cap T$, $S + T$"	$\dim S + \dim T = \dim(S \cap T) + \dim(S + T)$	7

Chapter 1 — Operations, monoids, groups (Seminar 1)

1.1 Operations and their laws

Method: an operation on A is a function $A \times A \to A$; the result must stay in A for every pair.

Ex. 1

Which ones of the usual symbols of addition, subtraction, multiplication and division define an operation (composition law) on the numerical sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$?

Ex. 2

Let $A = \{a_1, a_2, a_3\}$. Determine the number of: (i) operations on A; (ii) commutative operations on A; (iii) operations on A with identity element. Generalization for a set A with n elements ($n \in \mathbb{N}^*$).

1.2 Groups among the number sets

Ex. 3

Decide which ones of the numerical sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are groups together with the usual addition or multiplication.

1.3 Monoids and stable subsets

Method: associativity + identity element = monoid; a stable subset keeps the result inside.

Ex. 4

Let "$*$" be the operation defined on $\mathbb{R}$ by $x * y = x + y + xy$. Prove that: (i) $(\mathbb{R}, *)$ is a commutative monoid. (ii) The interval $[-1, \infty)$ is a stable subset of $(\mathbb{R}, *)$.

Ex. 5

Let "$*$" be the operation defined on $\mathbb{N}$ by $x * y = \gcd(x, y)$. (i) Prove that $(\mathbb{N}, *)$ is a commutative monoid. (ii) Show that $D_n = \{x \in \mathbb{N} \mid x \mid n\}$ $(n \in \mathbb{N}^*)$ is a stable subset of $(\mathbb{N}, *)$ and $(D_n, *)$ is a commutative monoid. (iii) Fill in the table of the operation "$*$" on D_6.

Ex. 6

Determine the finite stable subsets of $(\mathbb{Z}, \cdot)$.

1.4 Proofs in groups

Ex. 7

Let $(G, \cdot)$ be a group. Show that: (i) G is abelian $\iff \forall x, y \in G,\ (xy)^2 = x^2y^2$. (ii) If $x^2 = 1$ for every $x \in G$, then G is abelian.

Ex. 8

Let "$\cdot$" be an operation on a set A and let $X, Y \subseteq A$. Define an operation "$*$" on the power set $\mathcal{P}(A)$ by $X * Y = \{x \cdot y \mid x \in X,\ y \in Y\}$. Prove that: (i) If $(A, \cdot)$ is a monoid, then $(\mathcal{P}(A), *)$ is a monoid. (ii) If $(A, \cdot)$ is a group, then in general $(\mathcal{P}(A), *)$ is not a group.

Chapter 2 — Relations, functions, equivalences (Seminar 2)

2.1 Graphs of relations and counting

Ex. 1

Let r, s, t, v be the homogeneous relations defined on the set $M = \{2, 3, 4, 5, 6\}$ by $x\,r\,y \iff x < y$, $x\,s\,y \iff x \mid y$, $x\,t\,y \iff \gcd(x, y) = 1$, $x\,v\,y \iff x \equiv y \pmod 3$. Write the graphs R, S, T, V of the given relations.

Ex. 2

Let A and B be sets with n and m elements respectively ($m, n \in \mathbb{N}^*$). Determine the number of: (i) relations having the domain A and the codomain B; (ii) homogeneous relations on A.

2.2 Reflexivity, transitivity, symmetry

Method: to prove a property, take arbitrary elements; to disprove it, give one concrete counterexample.

Ex. 3

Give examples of relations having each one of the properties of reflexivity, transitivity and symmetry, but not the others.

Ex. 4

Which ones of the properties of reflexivity, transitivity and symmetry hold for the following homogeneous relations: the strict inequality relations on $\mathbb{R}$, the divisibility relation on $\mathbb{N}$ and on $\mathbb{Z}$, the perpendicularity relation of lines in space, the parallelism relation of lines in space, the congruence of triangles in a plane, the similarity of triangles in a plane?

2.3 Equivalence relations and partitions

Method: equivalence relations on M ↔ partitions of M; the blocks are the equivalence classes.

Ex. 5

Let $M = \{1, 2, 3, 4\}$, let r_1, r_2 be homogeneous relations on M and let π_1, π_2, where $R_1 = \Delta_M \cup \{(1, 2), (2, 1), (1, 3), (3, 1), (2, 3), (3, 2)\}$, $R_2 = \Delta_M \cup \{(1, 2), (1, 3)\}$, $\pi_1 = \{\{1\}, \{2\}, \{3, 4\}\}$, $\pi_2 = \{\{1\}, \{1, 2\}, \{3, 4\}\}$. (i) Are r_1, r_2 equivalences on M? If yes, write the corresponding partition. (ii) Are π_1, π_2 partitions on M? If yes, write the corresponding equivalence relation.

Ex. 6

Define on $\mathbb{C}$ the relations r and s by: $z_1 \, r \, z_2 \iff |z_1| = |z_2|$; $z_1 \, s \, z_2 \iff \arg z_1 = \arg z_2$ or $z_1 = z_2 = 0$. Prove that r and s are equivalence relations on $\mathbb{C}$ and determine the quotient sets (partitions) $\mathbb{C}/r$ and $\mathbb{C}/s$ (geometric interpretation).

Ex. 7

Let $n \in \mathbb{N}$. Consider the relation ρ_n on $\mathbb{Z}$, called the congruence modulo n, defined by $x \, \rho_n \, y \iff n \mid (x - y)$. Prove that ρ_n is an equivalence relation on $\mathbb{Z}$ and determine the quotient set (partition) $\mathbb{Z}/\rho_n$. Discuss the cases $n = 0$ and $n = 1$.

Ex. 8

Determine all equivalence relations and all partitions on the set $M = \{1, 2, 3\}$.

2.4 Functions and other relations

Ex. 9

Let $M = \{0, 1, 2, 3\}$ and let $h = (\mathbb{Z}, M, H)$ be a relation, where $H = \{(x, y) \in \mathbb{Z} \times M \mid \exists z \in \mathbb{Z} : x = 4z + y\}$. Is h a function?

Ex. 10

Consider the following homogeneous relations on $\mathbb{N}$, defined by: $m \, r \, n \iff \exists a \in \mathbb{N} : m = 2^a n$; $m \, s \, n \iff (m = n \text{ or } m = n^2 \text{ or } n = m^2)$. Are r and s equivalence relations?

Chapter 3 — Groups, rings, subgroups, homomorphisms (Seminar 3)

3.1 Groups and rings of functions

Ex. 1

Let M be a non-empty set and let $S_M = \{f : M \to M \mid f \text{ is bijective}\}$. Show that $(S_M, \circ)$ is a group, called the symmetric group of M.

Ex. 2

Let M be a non-empty set and let $(R, +, \cdot)$ be a ring. Define on $R^M = \{f \mid f : M \to R\}$ two operations by: $\forall f, g \in R^M$, $(f + g)(x) = f(x) + g(x)$, $(f \cdot g)(x) = f(x) \cdot g(x)$, $\forall x \in M$. Show that $(R^M, +, \cdot)$ is a ring. If R is commutative or has identity, does R^M have the same property?

3.2 Subgroups and subrings

Method: use the characterization theorems, never the definition directly.

Ex. 3

Prove that $H = \{z \in \mathbb{C} \mid |z| = 1\}$ is a subgroup of $(\mathbb{C}^*, \cdot)$, but not of $(\mathbb{C}, +)$.

Ex. 4

Let $U_n = \{z \in \mathbb{C} \mid z^n = 1\}$ ($n \in \mathbb{N}^*$) be the set of n-th roots of unity. Prove that U_n is a subgroup of $(\mathbb{C}^*, \cdot)$.

Ex. 5

Let $n \in \mathbb{N}$, $n \geq 2$. Prove that: (i) $\mathrm{GL}_n(\mathbb{C}) = \{A \in M_n(\mathbb{C}) \mid \det(A) \neq 0\}$ is a stable subset of the monoid $(M_n(\mathbb{C}), \cdot)$; (ii) $(\mathrm{GL}_n(\mathbb{C}), \cdot)$ is a group, called the general linear group of rank n; (iii) $\mathrm{SL}_n(\mathbb{C}) = \{A \in M_n(\mathbb{C}) \mid \det(A) = 1\}$ is a subgroup of the group $(\mathrm{GL}_n(\mathbb{C}), \cdot)$.

Ex. 6

Show that the following sets are subrings of the corresponding rings: (i) $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ in $(\mathbb{C}, +, \cdot)$. (ii) $M = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \middle| \, a, b, c \in \mathbb{R} \right\}$ in $(M_2(\mathbb{R}), +, \cdot)$.

3.3 Homomorphisms and isomorphisms

Ex. 7

(i) Let $f : \mathbb{C}^* \to \mathbb{R}^*$ be defined by $f(z) = |z|$. Show that f is a group homomorphism between $(\mathbb{C}^*, \cdot)$ and $(\mathbb{R}^*, \cdot)$. (ii) Let $g : \mathbb{C}^* \to \mathrm{GL}_2(\mathbb{R})$ be defined by $g(a + bi) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$. Show that g is a group homomorphism between $(\mathbb{C}^*, \cdot)$ and $(\mathrm{GL}_2(\mathbb{R}), \cdot)$.

Ex. 8

Let $n \in \mathbb{N}$, $n \geq 2$. Prove that the groups $(\mathbb{Z}_n, +)$ of residue classes modulo n and $(U_n, \cdot)$ of n-th roots of unity are isomorphic.

Ex. 10

Let $M = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \middle| \, a, b \in \mathbb{R} \right\} \subseteq M_2(\mathbb{R})$. Show that $(M, +, \cdot)$ is a field isomorphic to $(\mathbb{C}, +, \cdot)$.

3.4 Invertible elements of $\mathbb{Z}_n$

Ex. 9

Let $n \in \mathbb{N}$, $n \geq 2$. Consider the ring $(\mathbb{Z}_n, +, \cdot)$ and let $\widehat{a} \in \mathbb{Z}_n^*$. (i) Prove that $\widehat{a}$ is invertible $\iff$ $(a, n) = 1$. (ii) Deduce that $(\mathbb{Z}_n, +, \cdot)$ is a field $\iff$ n is prime.

Chapter 4 — Vector spaces and subspaces (Seminar 4)

4.1 Checking the vector space axioms

Method: $(V, +)$ must be an abelian group, then check (L1)–(L4). To show "not a vector space", one failing axiom with concrete numbers is enough.

Ex. 1

Let K be a field. Show that $K[X]$ is a K-vector space, where the addition is the usual addition of polynomials and the scalar multiplication is defined as follows: $\forall k \in K$, $\forall f = a_0 + a_1X + \cdots + a_nX^n \in K[X]$, $k \cdot f = (ka_0) + (ka_1)X + \cdots + (ka_n)X^n$.

Ex. 2

Let K be a field and $m, n \in \mathbb{N}$, $m, n \geq 2$. Show that $M_{m,n}(K)$ is a K-vector space, with the usual addition and scalar multiplication of matrices.

Ex. 3

Let K be a field, $A \neq \emptyset$ and denote $K^A = \{f \mid f : A \to K\}$. Show that K^A is a K-vector space, where $(f + g)(x) = f(x) + g(x)$ and $(k \cdot f)(x) = k \cdot f(x)$, $\forall x \in A$.

Ex. 4

Let $V = \{x \in \mathbb{R} \mid x > 0\}$ and define the operations $x \perp y = xy$ and $k \top x = x^k$, $\forall k \in \mathbb{R}$ and $\forall x, y \in V$. Prove that V is a vector space over $\mathbb{R}$.

Ex. 5

Let K be a field and let $V = K \times K$. Decide whether V is a K-vector space with respect to the following addition and scalar multiplication: (i) $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2,\ y_1 + 2y_2)$ and $k \cdot (x_1, y_1) = (kx_1, ky_1)$; (ii) $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2,\ y_1 + y_2)$ and $k \cdot (x_1, y_1) = (kx_1, y_1)$, $\forall (x_1, y_1), (x_2, y_2) \in V$ and $\forall k \in K$.

4.2 Vector spaces over $\mathbb{Z}_p$

Ex. 6

Let p be a prime number and let V be a vector space over the field $\mathbb{Z}_p$. (i) Prove that $\underbrace{x+\cdots+x}_{p \text{ times}} = 0$, $\forall x \in V$. (ii) Is there a scalar multiplication endowing $(\mathbb{Z},+)$ with a structure of a vector space over $\mathbb{Z}_p$?

4.3 Subspaces

Method: characterization of subspaces — $0 \in S$ and $k_1 v_1 + k_2 v_2 \in S$.

Ex. 7

Which ones of the following sets are subspaces of the real vector space $\mathbb{R}^3$: (i) $A = \{(x,y,z) \in \mathbb{R}^3 \mid x = 0\}$; (ii) $B = \{(x,y,z) \in \mathbb{R}^3 \mid x = 0 \text{ or } z = 0\}$; (iii) $C = \{(x,y,z) \in \mathbb{R}^3 \mid x \in \mathbb{Z}\}$; (iv) $D = \{(x,y,z) \in \mathbb{R}^3 \mid x+y+z = 0\}$; (v) $E = \{(x,y,z) \in \mathbb{R}^3 \mid x+y+z = 1\}$; (vi) $F = \{(x,y,z) \in \mathbb{R}^3 \mid x = y = z\}$?

Ex. 8

Which ones of the following sets are subspaces: (i) $[-1,1]$ of the real vector space $\mathbb{R}$; (ii) $\{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \leq 1\}$ of the real vector space $\mathbb{R}^2$; (iii) $\left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \middle| \, a,b,c \in \mathbb{Q} \right\}$ of ${}_{\mathbb{Q}}M_2(\mathbb{Q})$ or of ${}_{\mathbb{R}}M_2(\mathbb{R})$; (iv) $\{f : \mathbb{R} \to \mathbb{R} \mid f \text{ continuous}\}$ of the real vector space $\mathbb{R}^{\mathbb{R}}$?

Ex. 9

Which ones of the following sets are subspaces of the K-vector space $K[X]$: (i) $K_n[X] = \{f \in K[X] \mid \deg f \leq n\}$ $(n \in \mathbb{N})$; (ii) $K'_n[X] = \{f \in K[X] \mid \deg f = n\}$ $(n \in \mathbb{N})$?

Ex. 10

Show that the set of all solutions of a homogeneous system of two equations and two unknowns with real coefficients is a subspace of the real vector space $\mathbb{R}^2$.

Chapter 5 — Generated subspaces, sums, linear maps (Seminar 5)

5.1 Generated subspaces

Method: $\langle v_1, \ldots, v_n \rangle$ = all linear combinations $k_1 v_1 + \cdots + k_n v_n$.

Ex. 1

Determine the following generated subspaces: (i) $\langle 1, X, X^2 \rangle$ in the real vector space $\mathbb{R}[X]$. (ii) $\left\langle \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\rangle$ in the real vector space $M_2(\mathbb{R})$.

Ex. 2

Consider the following subspaces of the real vector space $\mathbb{R}^3$: (i) $A = \{(x, y, z) \in \mathbb{R}^3 \mid x = 0\}$; (ii) $B = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 0\}$; (iii) $C = \{(x, y, z) \in \mathbb{R}^3 \mid x = y = z\}$. Write A, B, C as generated subspaces with a minimal number of generators.

Ex. 3

Consider the following vectors in the real vector space $\mathbb{R}^3$: $a = (-2, 1, 3)$, $b = (3, -2, -1)$, $c = (1, -1, 2)$, $d = (-5, 3, 4)$, $e = (-9, 5, 10)$. Show that $\langle a, b \rangle = \langle c, d, e \rangle$.

5.2 Direct sums

Method: $V = S \oplus T \iff V = S + T$ and $S \cap T = \{0\}$.

Ex. 4

Let $S = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 0\}$, $T = \{(x, y, z) \in \mathbb{R}^3 \mid x = y = z\}$. Prove that S and T are subspaces of the real vector space $\mathbb{R}^3$ and $\mathbb{R}^3 = S \oplus T$.

Ex. 5

Let S and T be the set of all even functions and of all odd functions in $\mathbb{R}^{\mathbb{R}}$ respectively. Prove that S and T are subspaces of the real vector space $\mathbb{R}^{\mathbb{R}}$ and $\mathbb{R}^{\mathbb{R}} = S \oplus T$.

5.3 Linear maps

Method: $f(k_1 v_1 + k_2 v_2) = k_1 f(v_1) + k_2 f(v_2)$. If $f(0) \neq 0$, f is not linear.

Ex. 6

Let $f, g : \mathbb{R}^2 \to \mathbb{R}^2$ and $h : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by $f(x, y) = (x + y,\ x - y)$, $g(x, y) = (2x - y,\ 4x - 2y)$, $h(x, y, z) = (x - y,\ y - z,\ z - x)$. Show that $f, g \in \mathrm{End}_{\mathbb{R}}(\mathbb{R}^2)$ and $h \in \mathrm{End}_{\mathbb{R}}(\mathbb{R}^3)$.

Ex. 7

Which ones of the following functions are endomorphisms of the real vector space $\mathbb{R}^2$: (i) $f : \mathbb{R}^2 \to \mathbb{R}^2$, $f(x, y) = (ax + by,\ cx + dy)$, where $a, b, c, d \in \mathbb{R}$; (ii) $g : \mathbb{R}^2 \to \mathbb{R}^2$, $g(x, y) = (a + x,\ b + y)$, where $a, b \in \mathbb{R}$?

Ex. 8

Let $a \in \mathbb{R}$ and let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $f(x, y) = (x \cos a - y \sin a,\ x \sin a + y \cos a)$. Prove that $f \in \mathrm{End}_{\mathbb{R}}(\mathbb{R}^2)$.

5.4 Kernel, image, fixed points

Ex. 9

Determine the kernel and the image of the endomorphisms from Exercise 6.

Ex. 10

Let V be a vector space over K and $f \in \mathrm{End}_K(V)$. Show that the set $S = \{x \in V \mid f(x) = x\}$ of fixed points of f is a subspace of V.

Chapter 6 — Linear independence and bases (Seminar 6)

6.1 Linear (in)dependence

Method: n vectors in K^n are independent $\iff$ their determinant is $\neq 0$; otherwise solve $k_1 v_1 + \cdots + k_m v_m = 0$.

Ex. 1

Let $v_1 = (1, -1, 0)$, $v_2 = (2, 1, 1)$, $v_3 = (1, 5, 2)$ be vectors in the canonical real vector space $\mathbb{R}^3$. Prove that: (i) v_1, v_2, v_3 are linearly dependent and determine a dependence relationship. (ii) v_1, v_2 are linearly independent.

Ex. 2

Prove that the following vectors are linearly independent: (i) $v_1 = (1, 0, 2)$, $v_2 = (-1, 2, 1)$, $v_3 = (3, 1, 1)$ in ${}_{\mathbb{R}}\mathbb{R}^3$. (ii) $v_1 = (1, 2, 3, 4)$, $v_2 = (2, 3, 4, 1)$, $v_3 = (3, 4, 1, 2)$, $v_4 = (4, 1, 2, 3)$ in ${}_{\mathbb{R}}\mathbb{R}^4$.

Ex. 3

Let $v_1 = (1, a, 0)$, $v_2 = (a, 1, 1)$, $v_3 = (1, 0, a)$ be vectors in ${}_{\mathbb{R}}\mathbb{R}^3$. Determine $a \in \mathbb{R}$ such that the vectors v_1, v_2, v_3 are linearly independent.

Ex. 4

Let $v_1 = (1, -2, 0, -1)$, $v_2 = (2, 1, 1, 0)$, $v_3 = (0, a, 1, 2)$ be vectors in ${}_{\mathbb{R}}\mathbb{R}^4$. Determine $a \in \mathbb{R}$ such that the vectors v_1, v_2, v_3 are linearly dependent.

6.2 Bases and coordinates

Method: coordinates = the unique solution of $v = k_1 v_1 + \cdots + k_n v_n$.

Ex. 5

Let $v_1 = (1,1,0)$, $v_2 = (-1,0,2)$, $v_3 = (1,1,1)$ be vectors in ${}_{\mathbb{R}}\mathbb{R}^3$. (i) Show that the list (v_1, v_2, v_3) is a basis of the real vector space $\mathbb{R}^3$. (ii) Express the vectors of the canonical basis (e_1, e_2, e_3) of $\mathbb{R}^3$ as a linear combination of the vectors v_1, v_2 and v_3. (iii) Determine the coordinates of $u = (1,-1,2)$ in each of the two bases.

Ex. 6

Let $n \in \mathbb{N}^*$. Show that the vectors $v_1 = (1,\ldots,1,1)$, $v_2 = (1,\ldots,1,2)$, $v_3 = (1,\ldots,1,2,3)$, ..., $v_n = (1,2,\ldots,n-1,n)$ form a basis of the real vector space $\mathbb{R}^n$ and write the coordinates of a vector $(x_1,\ldots,x_n)$ in this basis.

Ex. 7

Let $E_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $E_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $E_3 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, $E_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, $A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $A_2 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, $A_3 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, $A_4 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. Prove that the lists (E_1, E_2, E_3, E_4) and (A_1, A_2, A_3, A_4) are bases of the real vector space $M_2(\mathbb{R})$ and determine the coordinates of $B = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$ in each of the two bases.

Ex. 8

Let $\mathbb{R}_2[X] = \{f \in \mathbb{R}[X] \mid \deg f \leq 2\}$. Show that the lists $E = (1, X, X^2)$, $B = (1,\ X-a,\ (X-a)^2)$ ($a \in \mathbb{R}$) are bases of the real vector space $\mathbb{R}_2[X]$ and determine the coordinates of a polynomial $f = a_0 + a_1 X + a_2 X^2 \in \mathbb{R}_2[X]$ in each basis.

6.3 Counting over $\mathbb{Z}_2$

Ex. 9

Determine the number of bases of the vector space $\mathbb{Z}_2^3$ over $\mathbb{Z}_2$.

Ex. 10

Determine the number of elements of the general linear group $(\mathrm{GL}_3(\mathbb{Z}_2), \cdot)$ of invertible 3×3-matrices over $\mathbb{Z}_2$.

Chapter 7 — Dimension (Seminar 7)

7.1 Basis and dimension of a subspace

Method: parametrize the subspace, read off generators, check that they are independent; dim = number of vectors.

Ex. 1

Determine a basis and the dimension of the following subspaces of the real vector space $\mathbb{R}^3$: $A = \{(x, y, z) \in \mathbb{R}^3 \mid z = 0\}$, $B = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 0\}$, $C = \{(x, y, z) \in \mathbb{R}^3 \mid x = y = z\}$.

Ex. 2

Let K be a field and $S = \{(x_1, \ldots, x_n) \in K^n \mid x_1 + \cdots + x_n = 0\}$. (i) Prove that S is a subspace of the canonical vector space K^n over K. (ii) Determine a basis and the dimension of S.

Ex. 3

Determine a basis and the dimensions of the vector spaces $\mathbb{C}$ over $\mathbb{C}$ and $\mathbb{C}$ over $\mathbb{R}$. Prove that the set $\{1, i\}$ is linearly dependent in the vector space $\mathbb{C}$ over $\mathbb{C}$ and linearly independent in the vector space $\mathbb{C}$ over $\mathbb{R}$.

7.2 Kernel and image

Method: $\dim V = \dim \operatorname{Ker} f + \dim \operatorname{Im} f$ (First Dimension Theorem) — use it to check your answers.

Ex. 4

Let $f : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $f(x, y, z) = (y, -x)$. Prove that f is an $\mathbb{R}$-linear map and determine a basis and the dimension of $\operatorname{Ker} f$ and $\operatorname{Im} f$.

Ex. 5

Let $f \in \operatorname{End}_{\mathbb{R}}(\mathbb{R}^3)$ be defined by $f(x, y, z) = (-y + 5z,\ x,\ y - 5z)$. Determine a basis and the dimension of $\operatorname{Ker} f$ and $\operatorname{Im} f$.

7.3 Completion to a basis and complements

Ex. 6

Complete the bases of the subspaces from Exercise 1 to some bases of the real vector space $\mathbb{R}^3$ over $\mathbb{R}$.

Ex. 7

Determine a complement for the following subspaces: (i) $A = \{(x, y, z) \in \mathbb{R}^3 \mid x + 2y + 3z = 0\}$ in the real vector space $\mathbb{R}^3$; (ii) $B = \{aX + bX^3 \mid a, b \in \mathbb{R}\}$ in the real vector space $\mathbb{R}_3[X]$.

7.4 Dimension formulas

Method: Second Dimension Theorem $\dim S + \dim T = \dim(S \cap T) + \dim(S + T)$.

Ex. 8

Let V be a vector space over K and let S, T and U be subspaces of V such that $\dim(S \cap U) = \dim(T \cap U)$ and $\dim(S + U) = \dim(T + U)$. Prove that if $S \subseteq T$, then $S = T$.

Ex. 9

Consider the subspaces $S = \{(x, y, z) \in \mathbb{R}^3 \mid x = 0\}$, $T = \langle (0, 1, 1), (1, 1, 0) \rangle$ of the real vector space $\mathbb{R}^3$. Determine $S \cap T$ and show that $S + T = \mathbb{R}^3$.

Ex. 10

Determine the dimensions of the subspaces S, T, $S + T$ and $S \cap T$ of the real vector space $M_2(\mathbb{R})$, where $S = \left\langle \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \right\rangle$, $T = \left\langle \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \right\rangle$.

Chapter 8 — True or false? (Crivei's chapter quizzes)

Based on the end-of-chapter quizzes of Crivei, *Basic Linear Algebra* (statements reformulated). Decide **T** or **F** and justify: a one-line proof for T, a concrete counterexample for F. A bare "T/F" earns nothing at the exam — the justification is the point.

8.1 Quiz 1 — Preliminaries (Chapters 1–3)

#	Statement	T / F
1	A symmetric homogeneous relation is always reflexive.	
2	The intersection of all the blocks of a partition of a set is always empty.	
3	A partition of a set A is a subset of A.	
4	A finite set has as many partitions as equivalence relations.	
5	Every group is a semigroup.	
6	In a non-abelian group, $xy \neq yx$ for all elements x, y.	
7	In every group, $(xy)^{-1} = x^{-1}y^{-1}$.	

8	A subgroup H of G has the same identity element as G.	
9	An element of a subgroup $H \leq G$ may have an inverse in H different from its inverse in G.	
10	Every subgroup of an abelian group is abelian.	
11	Every stable subset of a group is a subgroup.	
12	Between any two groups there is at least one group homomorphism.	
13	Every group homomorphism is an isomorphism.	
14	A ring is a group with respect to addition and also with respect to multiplication.	
15	If a ring R is not commutative, then $(R, +)$ is not commutative.	
16	In a commutative ring, $x^2 - y^2 = (x + y)(x - y)$ for all x, y.	
17	A field is an abelian group with respect to each of its two operations.	
18	Every field is a ring with identity.	
19	Every field is a division ring.	
20	A subring of a ring with identity has an identity element.	
21	Every subfield contains an identity element for multiplication.	
22	A ring homomorphism $R \to R'$ is a group homomorphism $(R, +) \to (R', +)$.	
23	A ring homomorphism between two unitary rings always maps 1 to $1'$.	
24	$\det(A + B) = \det A + \det B$ for all $A, B \in M_n(K)$.	
25	For $A \in M_{m,n}(K)$, $\operatorname{rank} A = \max(m, n)$.	

8.2 Quiz 2 — Vector spaces (Chapters 4–7)

#	Statement	T / F
1	The scalar multiplication of a K-vector space V is a function $V \times V \to V$.	
2	$(k_1 + k_2)v = k_2v + k_1v$ for all scalars k_1, k_2 and vectors v.	
3	$(k_1k_2)v = k_2(k_1v)$ for all scalars k_1, k_2 and vectors v.	
4	Every non-zero vector space has infinitely many vectors.	
5	Every non-zero vector space over an infinite field has infinitely many vectors.	
6	A subset of V containing the zero vector is a subspace.	
7	The intersection of any (non-empty) family of subspaces is a subspace.	
8	The union of any family of subspaces is a subspace.	
9	$\langle \emptyset \rangle = \emptyset$.	

10	A linear combination of linear combinations of some vectors is a linear combination of those vectors.	
11	For subspaces S, T: $S \cup T \subseteq S + T$.	
12	Every linear map is a group homomorphism.	
13	The inverse of a vector space isomorphism is linear.	
14	In a linearly dependent list, *every* vector is a linear combination of the others.	
15	If $v_1, \ldots, v_n$ are dependent, there are scalars $k_1, \ldots, k_n$, *all* non-zero, with $k_1 v_1 + \cdots + k_n v_n = 0$.	
16	If v_1, v_2 are independent and v_2, v_3 are independent, then v_1, v_3 are independent.	
17	Every vector space has a system of generators.	
18	If X generates V and one vector of X is a combination of the others, then the others still generate V.	
19	Every vector space has exactly one basis.	
20	The zero vector space has no basis.	
21	A vector space of dimension $n \geq 1$ can be generated by $n - 1$ vectors.	
22	A vector space of dimension n can contain $n + 1$ linearly independent vectors.	
23	Every subspace has exactly one complement.	
24	Every system of generators can be completed to a basis.	
25	For subspaces S, T: $\dim S + \dim T \leq \dim(S + T)$.	

Chapter 9 — Proof practice (exam style)

9.1 Course theorems — write the proof without looking

The exam asks for proofs of results from the course. Write each one on paper, then compare with *Solutions & Theory* (chapter in brackets). Tick it only when your proof has every step justified.

#	**Prove that…**	**Where**
1	the identity element is unique; the inverse of an element is unique (associative operation)	Ch. 1
2	$(xy)^{-1} = y^{-1}x^{-1}$ and the cancellation laws hold in a group	Ch. 1
3	equivalence relations on A correspond bijectively to partitions of A	Ch. 2
4	$H \leq G \iff 1 \in H$ and $xy^{-1} \in H$ for all $x, y \in H$	Ch. 3
5	a group homomorphism satisfies $f(1) = 1'$ and $f(x^{-1}) = f(x)^{-1}$	Ch. 3
6	$kv = 0 \iff k = 0$ or $v = 0$ (and the rules $k \cdot 0 = 0 \cdot v = 0$)	Ch. 4

7	$S \leq V \iff 0 \in S$ and $k_1 v_1 + k_2 v_2 \in S$	Ch. 4
8	$\langle X \rangle$ is the set of all linear combinations of vectors of X	Ch. 5
9	$V = S \oplus T \iff$ every vector is uniquely $s + t$	Ch. 5
10	$\operatorname{Ker} f$ and $\operatorname{Im} f$ are subspaces; $\operatorname{Ker} f = \{0\} \iff f$ injective	Ch. 5
11	B is a basis $\iff$ every vector has a unique writing in B	Ch. 6
12	Steinitz Exchange Theorem	Ch. 7
13	any two bases have the same number of vectors	Ch. 7
14	every independent list can be completed to a basis; every subspace has a complement	Ch. 7
15	First Dimension Theorem $\dim V = \dim \operatorname{Ker} f + \dim \operatorname{Im} f$	Ch. 7
16	Second Dimension Theorem $\dim S + \dim T = \dim(S \cap T) + \dim(S + T)$	Ch. 7

9.2 New proof exercises

Not from the seminar sheets — written in the style of the partial exam. Each uses one theorem from the course; full solutions in *Solutions & Theory*, Chapter 9.

P1

Let $(G, \cdot)$ be a group and $H, K \leq G$. (i) Prove that $H \cap K \leq G$. (ii) Prove that $H \cup K \leq G$ if and only if $H \subseteq K$ or $K \subseteq H$.

P2

Let $f : G \to G'$ be a group homomorphism and $\operatorname{Ker} f = \{x \in G \mid f(x) = 1'\}$. Prove that $\operatorname{Ker} f \leq G$, $\operatorname{Im} f \leq G'$, and that f is injective $\iff \operatorname{Ker} f = \{1\}$.

P3

Let $(G, \cdot)$ be a group. Prove that its *center* $Z(G) = \{a \in G \mid ax = xa,\ \forall x \in G\}$ is a subgroup of G.

P4

Let $(R, +, \cdot)$ be a unitary ring and $U(R)$ the set of its invertible elements. Prove that $(U(R), \cdot)$ is a group. Determine $U(\mathbb{Z})$ and $U(\mathbb{Z}_n)$.

P5

Prove that in a field, $xy = 0$ implies $x = 0$ or $y = 0$. Deduce that $\mathbb{Z}_6$ is not a field.

P6

On $\mathbb{Z} \times \mathbb{Z}^*$ define $(a,b) \sim (c,d) \iff ad = bc$. Prove that $\sim$ is an equivalence relation and describe its equivalence classes.

P7

Let S, T be subspaces of V. Prove that $S \cup T \leq V$ if and only if $S \subseteq T$ or $T \subseteq S$.

P8

Let $f : V \to V'$ be a linear map and $S' \leq V'$. Prove that $f^{-1}(S') = \{v \in V \mid f(v) \in S'\}$ is a subspace of V.

P9

Let $f \in \mathrm{End}_K(V)$ with $f \circ f = f$ (a *projection*). Prove that $V = \operatorname{Ker} f \oplus \operatorname{Im} f$.

P10

Let (v_1, v_2, v_3) be a basis of V. Prove that $(v_1,\ v_1 + v_2,\ v_1 + v_2 + v_3)$ is also a basis of V and find the coordinates of v_1, v_2, v_3 in it.

P11

Let $f : V \to V'$ be linear. Prove: if $\dim V > \dim V'$, then f is not injective; if $\dim V < \dim V'$, then f is not surjective.

P12

Let S, T be subspaces of V, $\dim V = n$. Prove that if $\dim S + \dim T > n$, then $S \cap T \neq \{0\}$. Deduce that any two planes through the origin in $\mathbb{R}^3$ intersect in at least a line.

P13

Let $f : V \to V'$ and $g : V' \to V''$ be linear. Prove that $\operatorname{Ker} f \subseteq \operatorname{Ker}(g \circ f)$ and $\operatorname{Im}(g \circ f) \subseteq \operatorname{Im} g$, and deduce that $\operatorname{rank}(g \circ f) \leq \min(\operatorname{rank} f, \operatorname{rank} g)$.